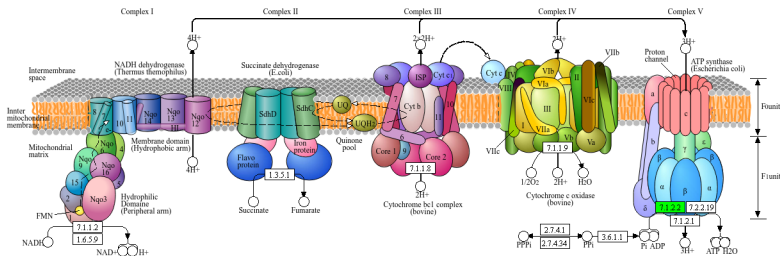
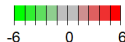




OXIDATIVE PHOSPHORYLATION



NADH dehydrogenase

E	ND1	ND2	ND3	ND4	ND4L	ND5	ND6													
E	Ndufa1	Ndufa2	Ndufa3	Ndufa4	Ndufa5	Ndufa6	Ndufa7	Ndufa8	Ndufv1	Ndufv2	Ndufv3									
B/A	NuoA	NuoB	NuoC	NuoD	NuoE	NuoF	NuoG	NuoH	NuoI	NuoJ	NuoK	NuoL	NuoM	NuoN						
E/B	NdhC	NdhK	NdhJ	NdhH	NdhA	NdhI	NdhG	NdhE	NdhF	NdhD	NdhB	NdhL	NdhM	NdhN	HoxE	HoxF	HoxU			
E	Ndufa1	Ndufa2	Ndufa3	Ndufa4	Ndufa5	Ndufa6	Ndufa7	Ndufa8	Ndufa9	Ndufa10	Ndufa11	Ndufa12	Ndufa13							
E	Ndufb1	Ndufb2	Ndufb3	Ndufb4	Ndufb5	Ndufb6	Ndufb7	Ndufb8	Ndufb9	Ndufb10	Ndufb11	Ndufb12	Ndufb13							

Succinate dehydrogenase / Fumarate reductase

E	SDHC	SDHD	SDHA	SDHB		
B/A	SdhC	SdhD	SdhA	SdhB		
			FrdA	FrdB	FrdC	FrdD

Cytochrome c reductase

E/B/A	ISP	Cytb	Cyt1																
E				COR1	QCR2	QCR6	QCR7	QCR8	QCR9	QCR10									

Cytochrome c oxidase

E	COX10	COX3	COX1	COX2	COX4	COX5A	COX5B	COX6A	COX6B	COX6C	COX7A	COX7B	COX7C	COX8	E/B/A	COX11	COX15	COX17
B/A	CyoE	CyoD	CyoC	CyoB	CyoA	CoxD	CoxC	CoxA	CoxB	QoxD	QoxC	QoxB	QoxA	SoxD	SoxC	SoxB	SoxA	

Cytochrome c oxidase, cbb3-type

B	I	II	IV	III
---	---	----	----	-----

Cytochrome bd complex

B/A	CydA	CydB	Cydc
-----	------	------	------

Cytochrome c

CYC

F-type ATPase (Bacteria)

alpha	beta	gamma	delta	epsilon
a	b	c		

F-type ATPase (Eukaryotes)

alpha	beta	gamma	delta	epsilon	
OSCP	a	b	c	d	e
f	g	85k	i	k	8

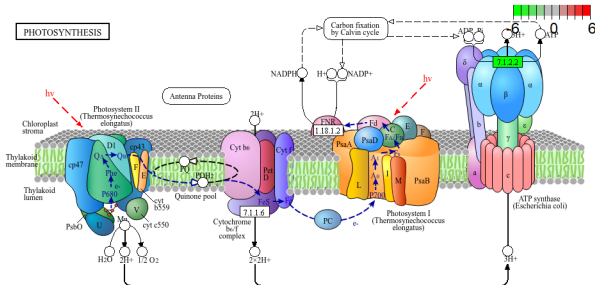
V/A-type ATPase (Bacteria, Archaea)

A	B	C	D	E	F	G/H
I	K					

V-type ATPase (Eukaryotes)

A	B	C	D	E	F	G	H
a	c	d	e	S1			

PHOTOSYNTHESIS



Photosystem II

D1	D2	cp43	cp47	cyt b559
PsbA	PsbD	PsbC	PsbB	PsbE
PsbF				

						MSP	OEC
PsbL	PsbJ	PsbK	PsbM	PsbH	PsbI	PsbO	PsbP
PsbQ	PsbR	PsbS	PsbT	PsbU	PsbV	PsbW	PsbX
PsbY	PsbZ	Psb27	Psb28	Psb28-2			

Photosystem I

PsaA	PsaB	PsaC	PsaD	PsaE	PsaF	PsaG	PsaH
PsaI	PsaJ	PsaK	PsaL	PsaM	PsaN	PsaO	PsaX

Cytochrome b₆/f complex

PetB	PetD	PetA	PetC	PetL	PetM	PetN	PetG
------	------	------	------	------	------	------	------

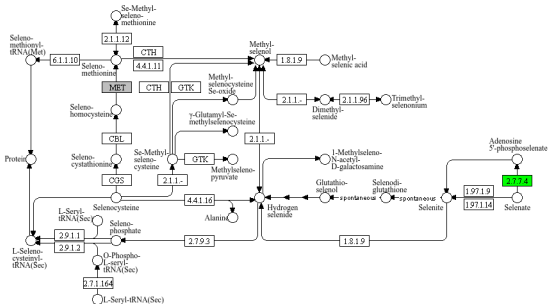
Photosynthetic electron transport

PC	Fd	FNR	cyt c6
PetE	PetF	PetH	PetU

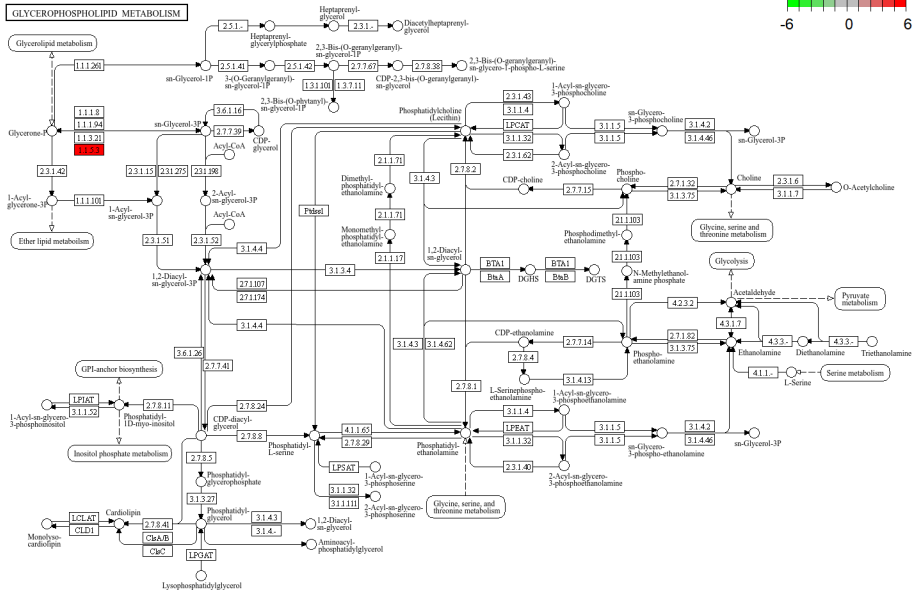
F-type ATPase

beta	alpha	gamma	delta	epsilon	c	a	b
------	-------	-------	-------	---------	---	---	---

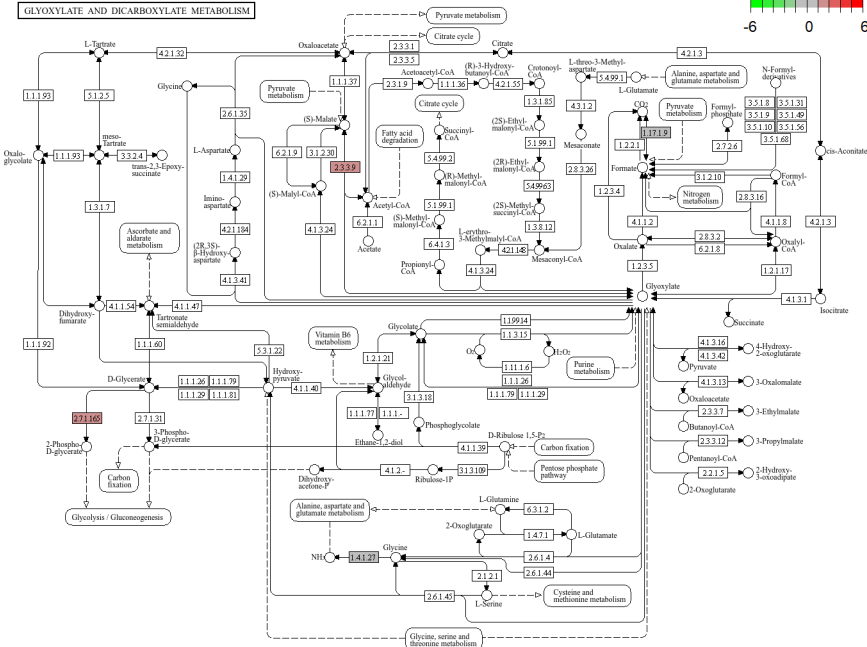
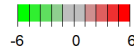
SELENOCOMPOUND METABOLISM



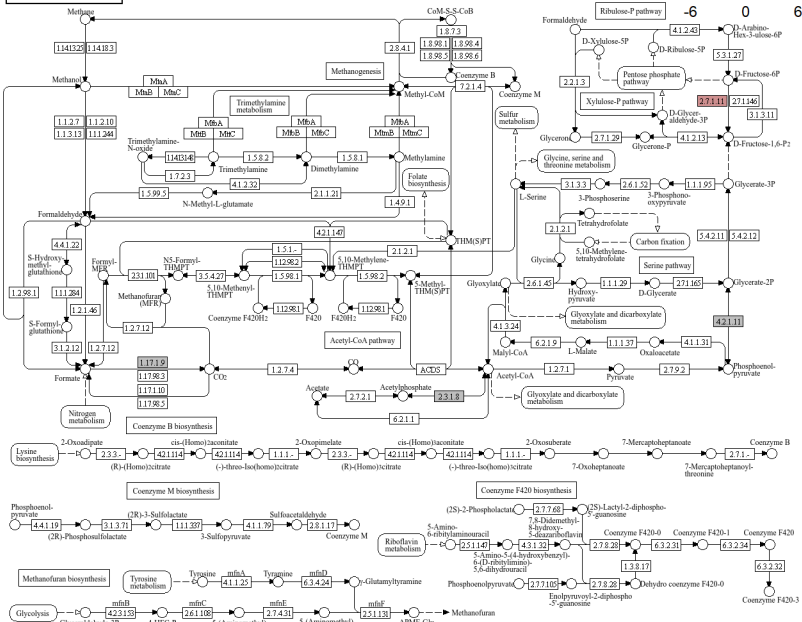




GLYOXYLATE AND DICARBOXYLATE METABOLISM

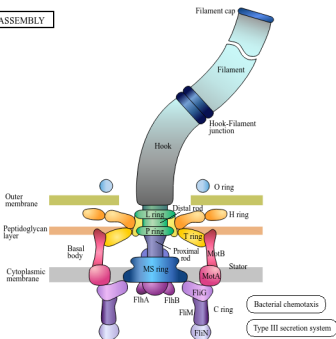


METHANE METABOLISM

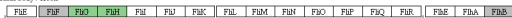




FLAGELLAR ASSEMBLY

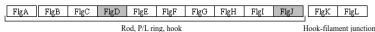


Basal body / Hook



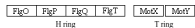
MS/C ring, Type III secretion system (T3SS)

T3SS



Rod, P/L ring, hook

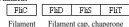
Hook-filament junction



H ring

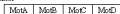
T ring

Filament

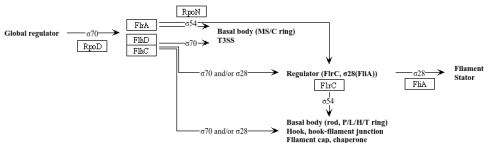


Filament Filament cap, chaperone

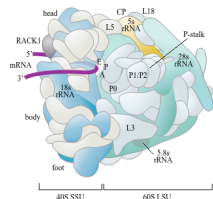
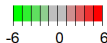
Stator



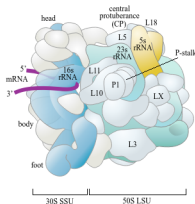
Regulator



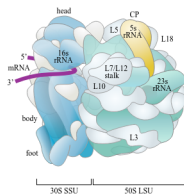
RIBOSOME



Eukaryotic 80S ribosome
(Homo sapiens)



Archaeal 70S ribosome
(Pyrococcus furiosus)



Bacterial 70S ribosome
(Escherichia coli)

Ribosomal RNAs

Eukaryotes 18S 28S/25S 5S 5.8S

Bacteria Archaea 16S 23S 5S

Mitochondria

12S 16S
18S/15S 26S/21S 5S

Chloroplast

16S 23S 5S 4.5S

Ribosomal proteins

Small subunit

S1e	S2e	S3e	S3Ae	S4e	S5e	S6e	S7e	S8e	S9e	S10e	S11e	S12e	S13e	S14e	S15e	S15Ae	S16e	S17e	S18e	S19e
S20e	S21e	S23e	S24e	S25e	S26e	S27e	S27Ae	S28e	S29e	S30e	RACK1	Asc1								
S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21
MRP518B	MRP522	MRP523	MRP524	MRP525	MRP526	MRP527	MRP528	MRP529	MRP530	MRP531	MRP533	MRP534	MRP535	MRP536	MRP10	MRP21	YMR31	GCD1		
PSRP2	PSRP3	PSRP4																		

Large subunit

L3e	L4e	L5e	L6e	L7e	L7Ae	L8e	L9e	L10e	L10Ae	L11e	L12e	L13e	L13Ae	L14e	L15e	L17e	L18e	L18Ae	LX	L19e
L21e	L22e	L23e	L23Ae	L24e	L26e	L27e	L27Ae	L28e	L29e	L30e	L31e	L32e	L34e	L35e	L35Ae	L36e	L37e	L37Ae	L38e	L39e
L40e	L41e	L44e	L36Ae	LP0	LP1	LP2	LP3													
L1	L2	L3	L4	L5	L6	L7A	L9	L10	L11	L7/L12	L12	L13	L14	L15	L16	L17	L18	L19	L20	L21
L22	L23	L24	L25	L27	L28	L29	L30	L31	L32	L33	L34	L35	L36							
MRPL37	MRPL38	MRPL39	MRPL40	MRPL41	MRPL42	MRPL43	MRPL44	MRPL45	MRPL46	MRPL47	MRPL48	MRPL49	MRPL50	MRPL51	MRPL52	MRPL53	MRPL54	MRPL55		
MRPL15	MRPL25	MRPL35	MRPL42	PSRP5	PSRP6															